M4. Reference Copies and a Shared Exam Hall Ticket

Week 3 OOP | Category B | Java solution with sample output

Objective

Implement the task using the classes, constructors, access control, references, or static members specified in the uploaded problem sheet.

Java Code

```
class HallTicket {
    String studentName;
    int seatNumber;

    HallTicket(String studentName, int seatNumber) {
        this.studentName = studentName;
        this.seatNumber = seatNumber;
    }
}

public class Main {
    public static void main(String[] args) {
        HallTicket priya = new HallTicket("Priya", 0);
        HallTicket copy = priya;       // Both variables refer to one object.
        copy.seatNumber = 45;

        HallTicket separate = new HallTicket("Priya", 45);

        System.out.println("Priya's seatNumber (via first variable):");
        System.out.println(priya.seatNumber);
        System.out.println("copy == priya: " + (copy == priya));
        System.out.println("separate == priya: " + (separate == priya));
    }
}
```

Sample Output

```
Priya's seatNumber (via first variable):
45
copy == priya: true
separate == priya: false
```

Concepts Covered

Reference copying, ==, object identity versus matching field values